

L -functions for all meromorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms, via manifestly finite contours

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Abstract

We give a single manifestly finite, regulator-free functional realizing in one object L -functions quasi-periods and period polynomials of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$, with generic poles throughout the upper half-plane $\mathbb{H}$ and at the cusp of $\mathrm{SL}_2(\mathbb{Z})$. Additionally, our construction preserves holomorphicity. It is a two-segment contour integral, $L^*(f, s) = L^*(f, s; \gamma_{\tau_0}^T) + L^*(f, s; \gamma_{\tau_0}^S)$, where the $\gamma_{\tau_0}^S$ is a path in $\mathbb{H}$ from $S(\tau_0)$ to τ_0 , and the same for $\gamma_{\tau_0}^T$. For weakly holomorphic forms our L -function is independent of τ_0 and of the path, and reproduces classical L -functions for holomorphic modular forms, reproduces the L -functions of weak modular forms in [1], which previously needed non-holomorphic regularization to be defined, and reproduces Brown's quasi-periods [2]. For properly meromorphic modular forms with poles inside the interior of $\mathbb{H}$, our L -function depends on the homotopy classes, defined in terms of pole-crossings. There is a natural choice of trivial homotopy class, which renders our L -functions canonical when the modular form has a pole in the interior of the fundamental domain, generalizing [3].

1 Introduction and preliminary concepts

Our primary goal in this note is to put forth a unifying generalization of the L -function associated to a given modular form. Our L -function naturally encompasses the classical case of L -functions associated to holomorphic modular forms [4], and seamlessly unifies Ref. [1]'s extension to L -functions of modular forms with poles at the cusp with Ref. [3]'s extension to L -functions of modular forms with poles at finite locations, exclusively away from cusps. Secondly, we show that our L -functions reproduce the numerical results for quasi-periods associated to weak cusp forms [2]/

1.1 Modular forms

Write $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ for the real and imaginary parts of a complex number $z \in \mathbb{C}$. Let $\mathbb{H} := \{\tau \in \mathbb{C} \mid \mathrm{Im}(\tau) > 0\}$ denote the upper half-plane. $\mathrm{SL}_2(\mathbb{Z})$ is the group freely generated by $T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

A *holomorphic modular form* of weight k for $\mathrm{SL}_2(\mathbb{Z})$ is a holomorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ such that $f(\tau+1) = f(-\frac{1}{\tau})\tau^{-k} = f(\tau)$ for all $\tau \in \mathbb{H}$, and $f(\tau) = \mathcal{O}(1)$ as $\mathrm{Im}(\tau) \rightarrow \infty$, and which has no poles at finite locations within $\mathbb{H}$. By virtue of the periodicity $f(\tau) = f(\tau+1)$ and holomorphicity, each holomorphic modular form has a q -series decomposition,

$$f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau} = \sum_n c_f(n) q^n \quad , \quad q := e^{2\pi i \tau} \quad . \quad (1)$$

In order for f to be bounded as $\tau \rightarrow i\infty$, we conclude that $c_f(n) = 0$ for all $n < 0$, i.e. that $f = \sum_{n \geq 0} c_f(n) q^n$; for such modular forms f we write $f \in M_k$. If a modular form f additionally has a vanishing constant term, i.e. if $c_f(n) = 0$, then f is a *cusp form*, and we write $f \in S_k$.

Clearly, $S_k \subset M_k$. Further, M_k is a finite-dimensional vector-space, as is S_k . Holomorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms must have even integer weights, and $k \geq 4$ [4].

A product of modular forms of weights k_1 and k_2 is a modular form of weight $k_1 + k_2$, so $M_* := \bigoplus_k M_k$ is naturally a (graded) ring. This graded ring is freely generated by the two modular forms $E_4 \in M_4$ and $E_6 \in M_6$, the $k = 4, 6$ specializations of the *weight k Eisenstein series* E_k , which may be defined as follows:

$$E_k := \sum_{(m,n)=1} \frac{1}{(m+n\tau)^k} = 1 + \frac{1}{\zeta(1-k)} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n, \quad (2)$$

where $\sum_{(m,n)=1}$ denotes summing over all relatively prime $(m,n) \in \mathbb{Z}^2$, $\zeta(s)$ is the Riemann zeta-function analytically continued to the entire s plane, and $\sigma_m(n) := \sum_{d|n} d^m$ is the Divisor function where $d|n$ means $n/d \in \mathbb{N}$. In short, $M_* = \mathbb{C}[E_4, E_6]$. We now define the *modular discriminant*:

$$\Delta := \frac{E_4^3 - E_6^2}{1728} = q + O(q^2) \in S_{12}. \quad (3)$$

Holomorphicity is most canonically relaxed via allowing a modular form to have a pole at the cusp of $\mathrm{SL}_2(\mathbb{Z})$, i.e. for a weight k modular form for $\mathrm{SL}_2(\mathbb{Z})$ to have negative powers of q in its q -series expansion. In this case, f diverges as $\tau \rightarrow i\infty$ as $f \sim e^{-2\pi N\tau}$, where $-N$ is the largest negative number such that $c_f(n)$ is non-zero. Such a modular form f is called a *weak modular form*, and we write $f \in M_k^!$. If a weak modular form f has a vanishing coefficient of q^0 , i.e. if $c_f(0) = 0$, it is called a *weak cusp form* and we write $f \in S_k^!$. Clearly, $S_k^! \subset M_k^!$, $S_k \subset S_k^!$ and $M_k \subset M_k^!$. Two important weak cusp forms in our paper are as follows:

$$J := \frac{E_4^3}{\Delta} - 744 = \frac{1}{q} + O(q) \in S_0^!, \quad (4)$$

$$\widehat{\Delta} = (J^2 + a_1 J + a_0) \Delta := \frac{1}{q} + O(q^2) \in S_{12}^!, \quad (5)$$

where a_0, a_1 are integers which exist to ensure the q -series gap that defines $\widehat{\Delta}$ [5].

In this work, we further relax from even weakly holomorphic modular forms, and consider *meromorphic modular forms* of weight k which are quotients: $f = \frac{f_1}{f_2}$ where $f_i \in M_{k_i}$, the denominator f_2 is not identically zero, and $k = k_1 - k_2$. We write F_k for the vector space of meromorphic modular forms of weight k , and set $F_* := \bigoplus_k F_k$. Then F_k vanishes unless k is an even integer, and F_* may be regarded as a “graded ring of fractions” of M_* .

1.2 L-functions for holomorphic modular forms and periods for cusp forms

For any $f \in S_k$, one may define the associated L-function:

$$L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s) := \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty f(it) t^{s-1} dt = \sum_{n \geq 1} \frac{c_f(n)}{n^s}. \quad (6)$$

A similar definition applies for $f \in M_k$. Similarly, one may define a period polynomial for $f \in S_k$ in terms of this L-function,

$$r_f(X, Y) := (2\pi i)^{k-1} \sum_{\ell=0}^{k-2} \binom{n}{\ell} \frac{L^*(f, \ell+1)}{i^{\ell+1}} X^{n-\ell} Y^\ell. \quad (7)$$

This polynomial, $r_f(X, Y) \in V_{k-2} := \mathbb{C}[X, Y]_{k-2}$, has a natural action under $\mathrm{SL}_2(\mathbb{Z})$, $(P|_\gamma)(X, Y) = P(aX + bY, cX + dY)$ for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. This period polynomial can be decomposed into a basis of highly constrained polynomials, which are annihilated by $|(1+T)$ and $|(1+U+U^2)$ [6], where $U := ST$ generates an order-3 cyclic subgroup of $\mathrm{SL}_2(\mathbb{Z})$. We define the *periods* of a cusp form to be the coefficients that localize r_f within this highly constrained sub-space of V_{k-2} .

However, as classically computed, these L-functions and associated period polynomials (when defined in terms of the special values of the associated L-function) are not defined when f has poles either at the cusp of $\mathrm{SL}_2(\mathbb{Z})$ or within the interior of $\mathbb{H}$. As stated above, there are extant, but partial and non-overlapping, extensions of the definition of $L(f, s)$ to $f \in M_k^!$ or $f \in F_k$. In this note, we provide a construction that unifies these disparate extensions.

2 A finite L-integral

Definition 2.1 (*S- and T-segments*). Let $\tau_0 \in \mathbb{H}$ and $f \in F_k$, let $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ denote the generators of $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, and let $f^{-1}(\infty)$ be the locus of all poles of $f \in \mathbb{H}$ (inclusive of the cusp of $\mathrm{SL}_2(\mathbb{Z})$). The S- and T-segments at τ_0 are smooth paths in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$,

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0, \quad (8)$$

running from $S^{-1}\tau_0 = -1/\tau_0$ and $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 .

Avoiding poles is automatic for $f \in M_k^!$, as long as the base-point τ_0 , is not at a cusp of $\mathrm{SL}_2(\mathbb{Z})$, i.e. there is only the trivial homotopy class. When $f \in F_k$, and f has poles at finite and non-zero imaginary heights, integration contours fall into nontrivial homotopy classes.

Definition 2.2 (*Hurwitz T-kernel*). Let $\zeta(s, x) = \sum_{m \geq 0} (m+x)^{-s}$ as a function $\zeta : \mathbb{C}_s \times (\mathbb{C} \setminus \mathbb{N})_x \rightarrow \mathbb{C}$ denote the Hurwitz zeta function (meromorphic in s , holomorphic in $x \in \mathbb{H}$). We define Hurwitz T-kernel as

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(1-(k-s), \tau+1). \quad (9)$$

Definition 2.3 (*L-integral*). Let $f \in F_k$, $\tau_0 \in \mathbb{H}$, and $s \in \mathbb{C}$. Further, let $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ be the S- and T-segments at τ_0 , as in Definition 2.1, and let $\tilde{k}_T$ be as defined in Definition 2.2. The L-integral of f is

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := e^{-\pi i s/2} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right]. \quad (10)$$

We use this L-integral to define a generalized L-function:

Definition 2.4 (*L-function*). Let $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ be as defined in 2.3. We define the generalized L-function of $f \in F_k$ to be,

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (11)$$

Having defined the L-integral and L-function, we prove they are manifestly finite for $f \in F_k$:

Lemma 2.5 (*Finiteness*). For $f \in F_k$, both integrals in Definition 2.3 are finite for every $s \in \mathbb{C}$, every $\tau_0 \in \mathbb{H}$, and every choice of paths $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$.

Proof. Both segments are compact and lie in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, so f is continuous on them; τ^{s-1} is continuous on $\mathbb{H}$, and $\tilde{k}_T(\tau, s)$ is continuous there since $\zeta(\nu, \tau+1)$ is holomorphic in $\tau \in \mathbb{H}$ (Definition 2.2). A continuous integrand on a compact path is integrable, for every $s \in \mathbb{C}$. $\square$

We now turn to path-dependence and τ_0 -independence of the L-integral and the L-function of $f \in F_k$. This holds for all $f \in M_k^!$:

Proposition 2.6 (Basepoint and path independence). *For $f \in M_k^!$ the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of $\tau_0 \in \mathbb{H}$ and of the paths: with no interior poles, $\mathbb{H}$ is simply connected, so all admissible contours are homotopic. We write $L^*(f, s)$ for the common value and $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$.*

Proof. With no interior poles the integrands $f(\tau)\tau^{s-1}$ and $f(\tau)\tilde{k}_T(\tau, s)$ are holomorphic on $\mathbb{H}$, which is simply connected; by Cauchy's theorem each integral is unchanged under any deformation fixing its endpoints, hence depends on those endpoints alone. This is path-independence.

For independence of τ_0 we differentiate the two integrals and watch them cancel. The T -segment runs from $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 — two endpoints a unit apart, moving together with τ_0 — so by Leibniz's rule and the T -periodicity $f(\tau_0 - 1) = f(\tau_0)$,

$$\partial_{\tau_0} \int_{\gamma^T} f \tilde{k}_T = f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) \delta_T \tilde{k}_T(\tau_0 - 1, s), \quad (12)$$

where $\delta_T \tilde{k}_T(\tau, s) := \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s)$. This is the whole role of δ_T : the unit gap between the endpoints turns the ∂_{τ_0} boundary term into a unit shift of the kernel. That shift is elementary — the kernel was built for it — since $\zeta(\nu, x + 1) - \zeta(\nu, x) = -x^{-\nu}$ gives, on both terms at $x = \tau + 1$,

$$\delta_T \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)}(\tau + 1)^{k-1-s} \quad (13)$$

$$\implies \partial_{\tau_0} \int_{\gamma^T} f \tilde{k}_T = f(\tau_0) [-\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s}]. \quad (14)$$

The S -segment runs $-1/\tau_0 \rightarrow \tau_0$; its lower endpoint moves with $\partial_{\tau_0}(-1/\tau_0) = \tau_0^{-2}$, so by Leibniz, the modularity $f(-1/\tau_0) = \tau_0^k f(\tau_0)$, and the phase $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{1-s}$,

$$\partial_{\tau_0} \int_{\gamma^S} f \tau^{s-1} = f(\tau_0) \tau_0^{s-1} - f(-1/\tau_0) (-1/\tau_0)^{s-1} \tau_0^{-2} = f(\tau_0) [\tau_0^{s-1} - e^{i\pi(s-1)} \tau_0^{k-1-s}]. \quad (15)$$

The two brackets are negatives of each other, so $\partial_{\tau_0} L^* = e^{-\pi i s/2} (\partial_{\tau_0} \int_{\gamma^S} + \partial_{\tau_0} \int_{\gamma^T}) = 0$. $\square$

Remark 2.7 (Classical L-function contour). *Let $f \in S_k$. As $\tau_0 \rightarrow i\infty$, the S -segment maps onto the standard L-function contour, $[0, i\infty]$. Simultaneously, as $f(it) \sim e^{-2\pi t}$ as $t \rightarrow \infty$, the T -segment vanishes exponentially. Thus, in this limit, $L^*(f, s)$ exactly maps onto the classical L-function integral [4]:*

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt. \quad (16)$$

3 L-functions for weak modular forms

Proposition 3.1 (Hurwitz-Gamma). *For $n \neq 0$, $\nu \in \mathbb{C}$, and $z \in \mathbb{H}$,*

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (17)$$

At $z = i$ the argument $-2\pi i n z$ collapses to $2\pi n$, giving $\Gamma(1-\nu, 2\pi n)$.

Proof. Write $\zeta(\nu, u) = \sum_{m \geq 0} (m+u)^{-\nu}$ and shift $u \mapsto u+m$ in the m -th term; as $e^{2\pi i n(u+m)} = e^{2\pi i n u}$ for integer m , the unit segments stack into one horizontal ray,

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = \sum_{m \geq 0} \int_{z+m}^{z+1+m} e^{2\pi i n v} v^{-\nu} dv = \int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv. \quad (18)$$

Substitute $w = -2\pi i n v$, so that $v = iw/(2\pi n)$, $dv = i dw/(2\pi n)$, and $e^{2\pi i n v} = e^{-w}$: the lower endpoint becomes $w = -2\pi i n z$, and for $n > 0$ the ray rotates to $w \rightarrow +\infty$ along the positive reals — legal, since $e^{-w} w^{-\nu}$ decays in the right half-plane, so the closing arc vanishes. With $v^{-\nu} = i^{-\nu} (2\pi n)^{\nu} w^{-\nu}$,

$$\int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv = i^{1-\nu} (2\pi n)^{\nu-1} \int_{-2\pi i n z}^{\infty} e^{-w} w^{-\nu} dw = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (19)$$

For $n < 0$ the ray instead grows as $\text{im } v \rightarrow \infty$ and the rotation is illegal; but both sides are analytic in n and agree for $n > 0$, so the identity continues to $n < 0$ with Γ on its principal branch. At $n = 0$ the prefactor $(2\pi n)^{\nu-1}$ is singular — that mode, the constant Fourier coefficient, is handled separately. $\square$

Theorem 3.2 (*L*-function for weakly holomorphic forms). *For $f \in M_k^!$, we may set $\tau_0 = i$, and prove*

$$L^*(f, s) = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right). \quad (20)$$

This L-integral satisfies the functional equation $L^(f, s) = i^k L^*(f, k-s)$.*

Proof. By path independence and base-point independence from Proposition 2.6, we may freely set $\tau_0 = i$. Here, the S -segment degenerates ($-1/i = i$), and the L -integral simplifies dramatically:

$$L^*(f, s) = e^{-\pi i s/2} \int_{i-1}^i f \tilde{k}_T d\tau. \quad (21)$$

We then expand f in terms of its q -series, $\sum_n c_f(n) e^{2\pi i n \tau}$, and substitute $u = \tau + 1$, and note that $e^{2\pi i n(u-1)} = e^{2\pi i n u}$. Here, each $n \neq 0$ mode is handled by Proposition 3.1. The first kernel term with $\nu = 1-s \leftrightarrow 1-\nu = s$ gives

$$e^{-\pi i s/2} e^{\pi i s/2} (2\pi n)^{-s} \Gamma(s, 2\pi n) = \Gamma(s, 2\pi n) / (2\pi n)^s, \quad (22)$$

and the second kernel term with $\nu = s-k+1 \leftrightarrow 1-\nu = k-s$ gives

$$-e^{-\pi i s/2} e^{i\pi(s-1)} e^{\pi i(k-s)/2} (2\pi n)^{s-k} \Gamma(k-s, 2\pi n) = i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (23)$$

So each $n \neq 0$ mode contributes $\Gamma(s, 2\pi n)/(2\pi n)^s + i^k \Gamma(k-s, 2\pi n)/(2\pi n)^{k-s}$. Crucially, the $n = 0$ mode contributes

$$c_f(0) e^{-\pi i s/2} \int_{i-1}^i \tilde{k}_T d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right). \quad (24)$$

Combining these reproduces Eq. (20). Finally, the functional equation, $L^*(f, s) = i^k L^*(f, k-s)$, follows by direct inspection. $\square$

3.1 Explicit S- and T-contours for general τ_0

Lemma 3.3 (*T-segment closed form on $M_k^!$*). For $f \in M_k^!$, for generic $\tau_0 \in \mathbb{H}$ away from cusps, denote the T-contour's contribution to $L^*(f, s)$ as $L_T^*(f, s; \gamma_{\tau_0}^T)$. For generic τ_0 , we have:

$$\begin{aligned} L_T^*(f, s; \gamma_{\tau_0}^T) = & -c_f(0) \left(\frac{e^{-\pi i s/2} \tau_0^s}{s} + \frac{e^{\pi i s/2} \tau_0^{k-s}}{k-s} \right) \\ & + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right). \end{aligned} \quad (25)$$

At $\tau_0 = i$ the constant mode collapses to $-c_f(0)(1/s + i^k/(k-s))$, recovering Theorem 3.2.

Proof. Expand $f = \sum_n c_f(n) q^n$ on the compact segment $[\tau_0 - 1, \tau_0]$, and restrict attention to the sub-sum, $\sum_{n \neq 0}$. For each such $n \neq 0$, apply the Proposition 3.1 in $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$ at $\nu = 1-s$ and $\nu = s-(k-1)$. The phases for the second term work-out identically as in the proof of Theorem 3.2.

For $n = 0$, the constant $c_f(0)$ multiplies $e^{-\pi i s/2} \int_{\tau_0-1}^{\tau_0} \tilde{k}_T(\tau, s) d\tau$; the Hurwitz antiderivative $\int_{\tau_0-1}^{\tau_0} \zeta(\sigma, \tau+1) d\tau = \tau_0^{1-\sigma}/(\sigma-1) - \text{from } \partial_\tau \zeta(\sigma-1, \tau+1) = -(\sigma-1)\zeta(\sigma, \tau+1) \text{ and the shift } \zeta(\sigma-1, \tau_0+1) - \zeta(\sigma-1, \tau_0) = -\tau_0^{1-\sigma} - \text{applied to the two halves of } \tilde{k}_T \text{ at } \sigma = 1-s \text{ and } \sigma = s-k+1 \text{ yields } -e^{-\pi i s/2} \tau_0^s/s \text{ and } -e^{\pi i s/2} \tau_0^{k-s}/(k-s) \text{ respectively.}$ $\square$

Corollary 3.4 (*T-segment cusp asymptotics*). Let $f \in S_k^!$, where $f = \sum_{n \geq -N} c_f(n) q^n$ for some $-N \in \mathbb{Z}$. The T-contour's contribution to $L^*(f, s)$ is given by Lemma 3.3. As $\tau_0 = i$ it goes to the cusp $i\infty$, the T-contour scales exactly as $f(it)$ scales: $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(it) \sim e^{2\pi N t}$.

Proof. For the asymptotics, $\Gamma(\sigma, z) \sim z^{\sigma-1} e^{-z}$ as $|z| \rightarrow \infty$ for $|\arg z| < \frac{3\pi}{2}$ [9, §8.11] gives $|c_f(n)|$ -weighted go as $e^{-2\pi n \operatorname{im}(\tau_0)}$. The smallest mode n in the q-series, i.e. $n = -N$, dominates the scaling of $f(it)$ and of $L_T^*(f, s; \gamma_{it}^T)$. Combining these results yields the claimed asymptotics. $\square$

Now, exploiting τ_0 -independence of $L^*(f, s)$, its explicit value as given in Theorem 3.2, and the explicit T-contribution $L_T^*(f, s; \gamma_{\tau_0}^T)$ trivially implies clean, closed-form expressions for the S-contour's contribution to $L^*(f, s)$:

Corollary 3.5 (*S-segment closed form*). For $f \in M_k^!$ and $\tau_0 \in \mathbb{H}$, the S-contribution $L_S^*(f, s; \gamma_{\tau_0}^S) := e^{-\pi i s/2} \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^{s-1} d\tau$ is the full L-value minus the T-contribution at the same base point, hence an explicit difference of incomplete gammas between the base points i and τ_0 :

$$L_S^*(f, s; \gamma_{\tau_0}^S) = L^*(f, s) - L_T^*(f, s; \gamma_{\tau_0}^T) \quad (26)$$

$$\begin{aligned} = & c_f(0) \left(\frac{e^{-\pi i s/2} \tau_0^s - 1}{s} + \frac{e^{\pi i s/2} \tau_0^{k-s} - i^k}{k-s} \right) \\ & + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, 2\pi n) - \Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n) - \Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right). \end{aligned} \quad (27)$$

Proof. Line (26) is the definition $L^*(f, s; \gamma_{\tau_0}^S) = L_S^*(\gamma_{\tau_0}^S) + L_T^*(\gamma_{\tau_0}^T)$ together with the τ_0 -independence of L^* (Proposition 2.6); evaluating at $\tau_0 = i$, where $L_S^*(\gamma_i^S) = 0$, identifies $L^*(f, s) = L_T^*(\gamma_i^T)$. Line (27) then inserts the closed forms of Theorem 3.2 for $L^*(f, s)$ and Lemma 3.3 for $L_T^*(\gamma_{\tau_0}^T)$: the $n \neq 0$ modes subtract base point by base point into the incomplete-gamma differences, and the two constant modes subtract to the $c_f(0)$ -terms in Eq. (27). $\square$

3.2 Classical Mellin recovery and canonical uniqueness.

As discussed in Remark 2.7, for $f \in S_k$, taking the cusp-limit $\tau_0 \rightarrow i\infty$ in Theorem 3.2 collapses the S -segment onto the classical Mellin contour $[0, i\infty]$ while the T -segment vanishes exponentially, since $f(it) = O(e^{-2\pi t})$ on the segment as $\tau = it \rightarrow i\infty$. I.e., here, one recovers the classical L-integral of f without obstruction:

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt. \quad (28)$$

Of course, path- and base-point independence of $L^*(f, s) = L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is guaranteed by Theorem 3.2 for all $f \in M_k^!$. However, these L-integrals are the sums of the S -contour and T -contour. Finiteness is ensured only of the sum of these two parts. Both contours may, and indeed do, grow without bound as $\tau_0 \rightarrow i\infty$ when $f \notin M_k$, as is highlighted in Corollary 3.4:

As $\tau_0 \rightarrow i\infty$ the polar mode q^{-N} of a properly weakly holomorphic modular form $f \in M_k^!$ makes the T -contribution grow like $e^{2\pi N \operatorname{Im} \tau_0}$ and the S -contribution diverge oppositely. Yet their sum, the τ_0 -independent value $L^*(f, s)$, remains finite and constant over all finite $\tau_0 \in \mathbb{H}$. Discarding $\gamma_{\tau_0}^T$ therefore does violence to the result: When the S -contour is isolated from its T -partner, and $\tau_0 \rightarrow i\infty$, it diverges. This is the precise reason that [1] needed to regularize their divergent L-integral representation for properly weak cusp forms.

3.3 Period polynomials, periods and quasi-periods

Because $L^*(f, s)$ is canonical for every $f \in M_k^!$, the period polynomial needs no separate construction — it is the generating polynomial of the critical values.

Definition 3.6 (Period polynomial). *For $f \in S_k^!$ and $n = k - 2$, the period polynomial is*

$$r_f(X, Y) := (2\pi i)^{n+1} \sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{n-\ell} Y^\ell. \quad (29)$$

Proposition 3.7 (Period polynomial as a contour integral). *With $(X - \tau Y)^n$ in place of the L -weight τ^{s-1} , and $\mathbf{K}_T(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \tilde{k}_T(\tau, \ell+1) X^{n-\ell} Y^\ell$ in place of $\tilde{k}_T$, the same two segments assemble r_f directly:*

$$r_f = (2\pi i)^{n+1} \left[\int_{\gamma^S} f(\tau) (X - \tau Y)^n d\tau + \int_{\gamma^T} f(\tau) \mathbf{K}_T(\tau; X, Y) d\tau \right]. \quad (30)$$

Proof. By Definition 2.3, the $X^{n-\ell} Y^\ell$ coefficient of the bracket is

$$r_f|_{X^{n-\ell} Y^\ell} = (-1)^\ell \binom{n}{\ell} \left[\int_{\gamma^S} f \tau^\ell + \int_{\gamma^T} f \tilde{k}_T(\tau, \ell+1) \right] = (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1). \quad (31)$$

This matches definition 3.6 term by term, and proves the claim. $\square$

Quasi-periods and their determinant: As an aside, the Hurwitz T -kernel in this note was originally uncovered in order to reproduce the manifestly finite computations of the quasi-periods for $\hat{\Delta} = 1/q + O(q^2) \in S_{12}^!$ originally given in [2]. In this vein, it is interesting to show collect quasi-periods computable from our L-integrals/period-polynomials for $f \in S_k^!$ (when $\dim(S_k) = 1$).

To this effect, we recall the notation of [5]. Decomposing the modular weight $k = 12\lambda + k'$ where $k' \in \{0, 4, 6, 8, 10, 14\}$, we see $\lambda = \dim(S_k)$ when $\lambda \geq 0$. For any λ , we may denote

$f_{k,m} := 1/q^m + O(q^{\lambda+1}) \in M_k^!$. When $\dim(S_k) = 1$, we have $f_{k,m} = 1/q^m + O(q^2)$. Here, define $\Delta_k := f_{k,-1} = q + O(q^2) \in S_k$ and $\widehat{\Delta}_k := f_{k,1} = 1/q + O(q^2) \in S_k^!$. Further, define

$$\text{pd}_k := \begin{pmatrix} \omega^+(\Delta_k) & \omega^-(\Delta_k) \\ \eta^+(\widehat{\Delta}_k) & \eta^-(\widehat{\Delta}_k) \end{pmatrix} \quad (32)$$

That $\det(\text{pd}_k)$ satisfies a Legendre-type relation, $\det(\text{pd}_k) \in \mathbb{Q}^\times \cdot (2\pi i)^{k-1}$, is the de Rham theory of Brown–Hain [7, Prop. 5.6]. Brown [2, §8.3] evaluates the constant at $k = 12$, finding $\det(\text{pd}_k) = -(k-2)! (2\pi i)^{k-1}$ — his period matrix orders the quasi-periods first, giving $+(k-2)!$, where ours orders the periods first — and remarks that he could locate no reference for the value. We reproduce $k = 12$ and give the analogous rational multiples for $k \in \{16, 18, 20, 22, 26\}$ in Table 1, below:

k	$\omega^+(f_k)$	$\omega^-(f_k)/i$	$\eta^+(\widehat{\Delta}_k)$	$\eta^-(\widehat{\Delta}_k)/i$	$\frac{-i^k \det(\text{pd}_k)}{(2\pi)^{k-1}(k-2)!}$
12	-6.89168e7	-5.58502e6	1.27202e11	1.02767e10	1
16	-8.81481e11	-1.54063e10	4.14120e13	6.83541e11	13/30
18	-5.07833e13	-5.43305e12	-2.19200e14	-3.36339e13	2/3
20	-4.22882e16	-3.73170e14	6.38640e16	-1.89823e14	17/5
22	-1.49974e18	-4.24489e16	1.43795e18	-4.41050e16	19/21
26	-1.18673e22	-3.60961e21	1.18653e22	-3.61016e21	23/15

Table 1: Periods $\omega^\pm(\Delta_k)$ and quasi-periods $\eta^\pm(\widehat{\Delta}_k)$, and the rational number $(-1)^{1+k/2} \det / ((2\pi i)^{k-1}(k-2)!)$ with $\det(\text{pd}_k) = \omega^+ \eta^- - \omega^- \eta^+$. At $k = 12$ the value matches Brown [2]; the entries for $k \geq 16$ appear to be new. These ratios are not intrinsic to f : they depend on how the rational period-polynomial basis $P^\pm$ is normalized (we use a primitive-integer normalization; Brown’s normalization gives $-(k-2)!$ at $k = 12$). We have not found a closed form for them. Full 50-digit values in [8].

4 Meromorphic modular forms

We find it convenient to define the following function,

$$G_{s,k}(n) := \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (33)$$

Proposition 4.1 (Basepoint and path independence within a homotopy class). *For $f \in F_k$, the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of τ_0 and of the path so long as both vary within a single homotopy class of admissible contours in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$. For $f \in M_k^!$ there is a single class, recovering Proposition 2.6; across different classes the value changes by the residues the contour sweeps, as quantified in Corollary 4.9.*

Proof. The τ_0 -derivative of Proposition 2.6 uses only the T -periodicity $f(\tau_0 - 1) = f(\tau_0)$ and the modularity $f(-1/\tau_0) = \tau_0^k f(\tau_0)$, both valid for every $f \in F_k$, and is evaluated at the moving endpoints alone. While τ_0 stays in one chamber of $\mathbb{H} \setminus f^{-1}(\infty)$ — where the segments meet no pole — the boundary terms cancel exactly as there, so $\partial_{\tau_0} L^* = 0$; the interior poles lie off the contour and never enter. Within a fixed class the integrands $f \tau^{s-1}$ and $f \tilde{k}_T$ are holomorphic along the contours, so any two are joined by a deformation avoiding $\Gamma \cdot f^{-1}(\infty)$, and Cauchy’s theorem equates their integrals. $\square$

Definition 4.2 (Polylog projection). *Let $f \in F_k$ have interior poles at the points τ_p of a set $\mathcal{P}_f$ of Γ -inequivalent representatives, and write $\text{Li}_s(z) = \sum_{n>0} z^n n^{-s}$. Since $\text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$ has a*

pole of order m at τ_p , the principal part of f there is matched by a unique polylog projection

$$\widehat{P}_{\tau_p} f(\tau) := \sum_{m>0} r_{f,\tau_p}^*(m) \operatorname{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)}), \quad (34)$$

its coefficients $r_{f,\tau_p}^*(m)$ being the principal-part data of f at τ_p . The projection $\widehat{P} := \sum_{\operatorname{Im} \tau_p \geq 1} \widehat{P}_{\tau_p}$ removes the principal part of every pole on or above the line $\operatorname{Im} \tau = 1$.

Definition 4.3 (Regularized cusp coefficients). With $\widehat{P}$ as in Definition 4.2,

$$c_{Rf}(n) := [(I - \widehat{P})f]_{q^n}. \quad (35)$$

Definition 4.4 (The space $F(\infty)$). $F(\infty)$ is the space of meromorphic g on $\mathbb{H}$ with $g(\tau) = g(\tau+1)$, finitely many poles in any compact subset of the strip $\{\operatorname{Im} \tau > 0, |\operatorname{Re} \tau| \leq \frac{1}{2}\}$, and $g(\tau) = O(e^{C \operatorname{Im} \tau})$ as $\operatorname{Im} \tau \rightarrow \infty$ for some C . Each such g has a Fourier expansion $g = \sum_n c_g(n) q^n$.

Then $F_k \subset F(\infty)$, and $(I - \widehat{P})f \in F(\infty)$ — the polylog tails are T -periodic but not modular, so the projection leaves F_k .

Lemma 4.5 (q -series-integral commutation). Let $g \in F(\infty)$ have no pole on or above the line $\operatorname{Im} \tau = 1$ (e.g. $g = (I - \widehat{P})f$). Then $g = \sum_n c_g(n) q^n$ converges along the T -segment at $\tau_0 = i$ and integrates term by term:

$$\int_{\gamma_i^T} g(\tau) \widetilde{k}_T(\tau, s) d\tau = \sum_n c_g(n) \int_{i-1}^i e^{2\pi i n \tau} \widetilde{k}_T(\tau, s) d\tau. \quad (36)$$

A pole at $\operatorname{Im} \tau_p \geq 1$ defeats this — there the coefficients grow at least like $n^{m-1} e^{2\pi n \operatorname{Im} \tau_p}$ [3, Lem. 2.6].

Proof. With no pole on or above $\operatorname{Im} \tau = 1$, the finitely many poles of g in the strip have heights < 1 ; let $h < 1$ be the largest. The Fourier series $g = \sum_n c_g(n) q^n$ converges absolutely and uniformly on compact subsets of $\{\operatorname{Im} \tau > h\}$, in particular on the T -segment $[i-1, i]$; with $\widetilde{k}_T(\tau, s)$ bounded there, the sum passes through the integral, which is the displayed identity. Conversely a pole at height $\operatorname{Im} \tau_p \geq 1$ forces $|c_g(n)| \gtrsim n^{m-1} e^{2\pi n \operatorname{Im} \tau_p}$ [3, Lem. 2.6], so $|c_g(n) q^n| \gtrsim n^{m-1} e^{2\pi n (\operatorname{Im} \tau_p - 1)} \not\rightarrow 0$ on $[i-1, i]$ and the series diverges. $\square$

Definition 4.6 (Polylog integral). For $N \geq 0$ and $z \in \mathbb{H}$, the polylog integral is the L -integral of a single tail $\operatorname{Li}_{-N}(e^{2\pi i(\tau-z)})$ along the T -segment at $\tau_0 = i$ (where γ^S degenerates, since $-1/i = i$):

$$I_N(s, z) := e^{-\pi i s/2} \int_{i-1}^i \operatorname{Li}_{-N}(e^{2\pi i(\tau-z)}) \widetilde{k}_T(\tau, s) d\tau. \quad (37)$$

Lemma 4.7 (Specialized polylog integral). Recall $G_{s,k}(n)$ from Eq. (33). For the projected poles $\operatorname{Im} z > 1$, where $|e^{2\pi i(\tau-z)}| \geq 1$ on the contour, polylog inversion $\operatorname{Li}_{-N}(w) = (-1)^{N+1} \operatorname{Li}_{-N}(1/w)$ resums the integral into the convergent series

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n z} G_{s,k}(-n). \quad (38)$$

This sum converges exponentially quickly, with base $e^{-2\pi n(y-1)}$, where $y = \operatorname{Im}(z)$.

Proof. For $\text{Im } z > 1$, $|e^{2\pi i(\tau-z)}| \geq 1$ on $[i-1, i]$, so the series of $\text{Li}_{-N}(e^{2\pi i(\tau-z)})$ diverges; the inversion rewrites it through $\text{Li}_{-N}(e^{2\pi i(z-\tau)}) = \sum_{n \geq 1} n^N e^{2\pi i n(z-\tau)}$, convergent since $|e^{2\pi i(z-\tau)}| \leq 1$. Integrating term by term against $\tilde{k}_T$, each $e^{-\pi i s/2} \int_{i-1}^i e^{-2\pi i n \tau} \tilde{k}_T d\tau$ is the building block $G_{s,k}(n)$ of Theorem 3.2 at mode $-n$, giving the displayed sum; for $N = 0$ the leftover -1 in $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$ adds $-e^{-\pi i s/2} \int_{i-1}^i \tilde{k}_T = \frac{1}{s} + \frac{i^k}{k-s}$.

Exponential convergence $\sim e^{-2\pi n(y-1)}$ for $y = \text{Im}(z) > 1$ follows from combining the asymptotic scaling $\Gamma(s, -2\pi n) \sim (2\pi n)^{s-1} e^{2\pi n}$ and the exact result $|e^{2\pi i n z}| = e^{-2\pi n y}$. $\square$

Theorem 4.8 (*L*-function of a meromorphic modular form). *For $f \in F_k$ with no pole on the edge $\text{Im } \tau = 1$, at $\tau_0 = i$,*

$$L^*(f, s) = -c_{Rf}(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}(n) + \sum_{\text{Im } \tau_p > 1} \sum_{m > 0} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p), \quad (39)$$

where, again $G_{s,k}(n)$ is the sum of incomplete gammas defined in Eq. (33); the regularized cusp sum (Definition 4.3, Theorem 3.2) plus one polylog integral (Lemma 4.7) per interior pole above the edge. A pole on the edge $\text{Im } \tau = 1$ — the elliptic point $\tau = i$ included — instead sits on the contour: there L^* is the principal value, and sweeping the contour across the pole changes it by the residue of Corollary 4.9.

Proof. At $\tau_0 = i$ the S -segment degenerates, so $L^*(f, s) = e^{-\pi i s/2} \int_{i-1}^i f \tilde{k}_T d\tau$. Split $f = (I - \hat{P})f + \hat{P}f$. The regularized part has no pole on or above $\text{Im } \tau = 1$ (Definition 4.2), so by Lemma 4.5 its Fourier series integrates term by term; mode by mode this is Theorem 3.2, giving $-c_{Rf}(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}(n)$. The projected part $\hat{P}f = \sum_{\tau_p} \sum_{m > 0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$ integrates to $\sum_{\text{Im } \tau_p > 1} \sum_{m > 0} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p)$ by Lemma 4.7. Their sum is the stated formula. $\square$

Corollary 4.9 (Wall-crossing and the winding data X_S, X_T). *Fix $f \in F_k$ and record, for an admissible pair $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$, its winding data — functions $X_S, X_T : \Gamma \cdot f^{-1}(\infty) \rightarrow \mathbb{Z}$ giving the net number of turns of γ^S , resp. γ^T , about each pole relative to a fixed reference pair (value $L_0^*(f, s)$ there). By Proposition 4.1 the integral depends only on (X_S, X_T) :*

$$L^*(f, s; X_S, X_T) = L_0^*(f, s) + 2\pi i e^{-\pi i s/2} \left[\sum_{\tau_p} X_S(\tau_p) \text{Res}_{\tau=\tau_p}(f \tau^{s-1}) + \sum_{\tau_p} X_T(\tau_p) \text{Res}_{\tau=\tau_p}(f \tilde{k}_T) \right]. \quad (40)$$

A single wall-crossing is the rank-one case: sweeping one segment once across one pole τ_p shifts L^* by $2\pi i e^{-\pi i s/2} \text{Res}_{\tau_p}(f \tau^{s-1})$ for γ^S , or $2\pi i e^{-\pi i s/2} \text{Res}_{\tau_p}(f \tilde{k}_T)$ for γ^T . Thus L^* is fixed by the contour only up to the residues its two segments wind around — weighted by τ^{s-1} on γ^S and by $\tilde{k}_T$ on γ^T .

Proof. Within-class constancy is Proposition 4.1. Two pairs whose winding data differ by one unit at a single τ_p on one segment bound a contour encircling τ_p once; the residue theorem adds $2\pi i$ times the residue of that segment's integrand — $f \tau^{s-1}$ for γ^S , $f \tilde{k}_T$ for γ^T — and the prefactor $e^{-\pi i s/2}$ of Definition 2.3 multiplies through. Summing over all crossings gives (40). $\square$

Remark 4.10 (The S -segment does not drop out). *For $f \in M_k^!$ the interior is pole-free, so $X_S \equiv X_T \equiv 0$ and L^* is path-independent (Proposition 2.6). For $f \in F_k$ both windings are live, and the S -segment is no exception. Its endpoints $-1/\tau_0$ and τ_0 are S -equivalent, so γ^S can be closed up through the group action into a loop — but a loop enclosing interior poles does not integrate*

to zero in the meromorphic case; it equals $2\pi i \sum(\text{residues})$. The identification of the S -endpoints therefore does not make γ^S negligible: it carries genuine residue data X_S , weighted by τ^{s-1} , on exactly the same footing as the T -segment's X_T weighted by $\tilde{k}_T$.

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